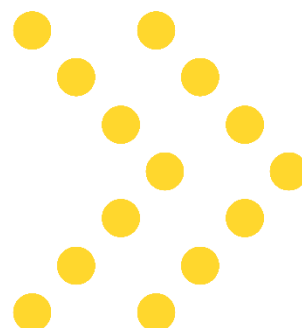


SKIDATA sweb Interfaces(R) 'Dynamic Data' REST

sweb® Interfaces 'Dynamic Data' (REST) API

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Version History

Date	Version	Changed by	Description of Changes
2022-01-12	1.0.0	Anders Malmberg	Initial Version.
2022-07-11	1.3	Anders Malmberg	Updated documentation for push notifications.
2022-08-09	1.4	Anders Malmberg	Updated documentation for push notifications Rename the artifact to sweb-Interface-DynamicData-REST-API.
2022-10-05	1.5	Anders Malmberg	Add an endpoint to query the carparks configured to trigger push notifications and

Date	Version	Changed by	Description of Changes
			one to return the integrator push notification config, e.g. URL and username.
2022-10-12	1.6	Anders Malmborg	Improved documentation for implementing the postEvent push API. Change `ce-dataschema` to `https://nexus.ski-data.net/nexus/repository/site/sweb-Interface-DynamicData-REST-API/doc/rest/dynamicdata.events/v1/dynamicdata.events.yaml`
2022-11-11	1.7	Anders Malmborg	Improved documentation for sequence of calls and integration tasks after feedback from pilot integrator.
2022-11-18	1.7.1	Anders Malmborg	Added internal link to trigger events.
2023-09-27	1.8	Anders Malmborg	Improve/clarify the description after feedback from integrators.
2024-04-29	1.8.1	Anders Malmborg	Align URLs in the samples, adapt HTTP examples to fit one page.

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1 Introduction

**Note:**

sweb® Interfaces is an API Gateway between an integrator system and SKIDATA systems, e.g. Parking.Logic.

This document describes sweb® Interface 'Dynamic Data' (REST) API. The API provides pull requests to get the counting areas and categories as well as push events to receive updates of them.

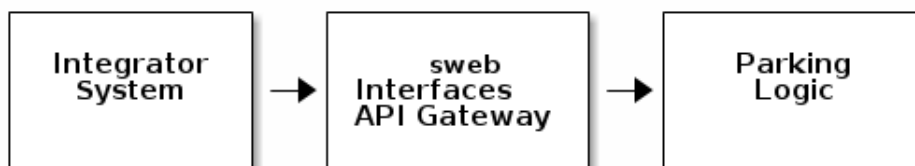
The API is a successor of the SOAP based sweb® Interface 'Dynamic Data' API.

Update of the counters, i.e. push events, are implemented using [HTTP Protocol Binding for CloudEvents - Version 1.0.2](#).

For all (upcoming) APIs that provide push events the endpoint implemented by the integrator will receive events to the same endpoint.

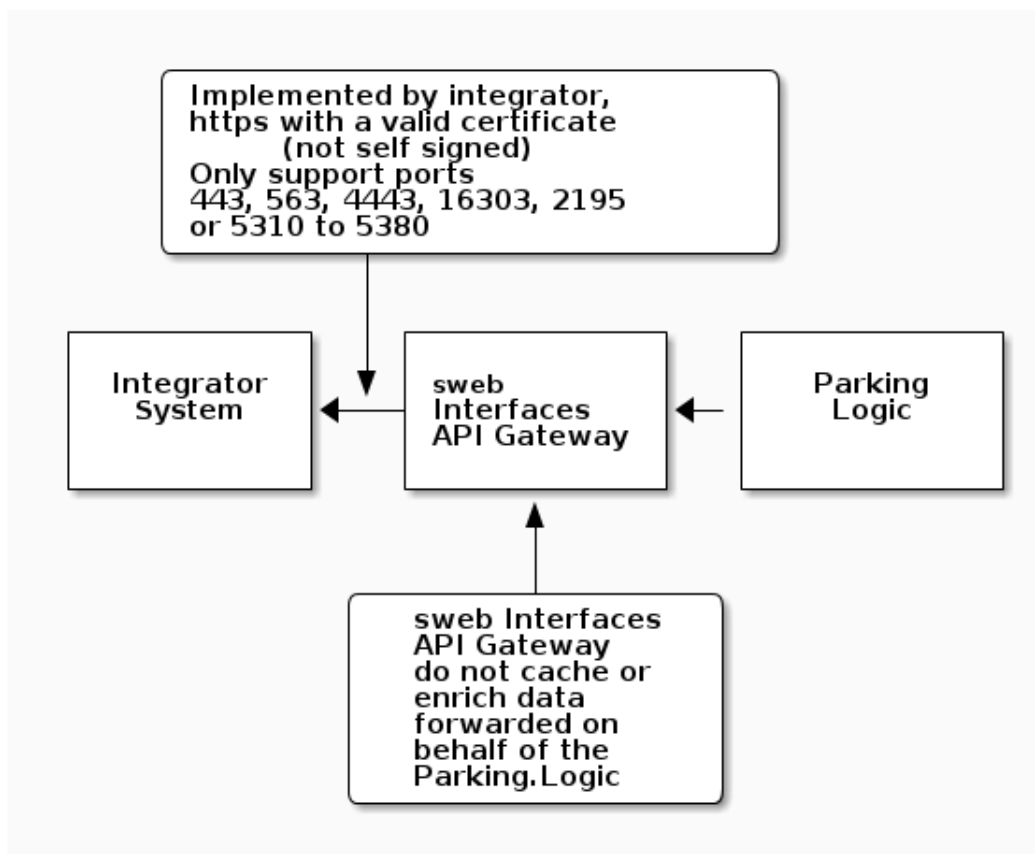
1.1 Integrator triggered configuration of the events

The integrator system implements [OpenAPI Definition for sweb® Interface 'Dynamic Data' \(REST\) API](#) as a client.



1.2 Integrator receiving the events

The integrator system implements [OpenAPI Definition for sweb® Interface 'Dynamic Data' \(REST\) Event API](#) as a server. Events are pushed to the integrator system based on the configuration with the [OpenAPI Definition for sweb® Interface 'Dynamic Data' \(REST\) API](#).



The event API is to be implemented by the **integrator**. The event API will be called for every update of a counting area/category.

1.3 Integration Roadmap

Note:



The SKIDATA Staging and Production environment are different cloud environments. A Parking.Logic can only be registered on one cloud environment. Consequently, to integrate with the SKIDATA Staging environment, a test Parking.Logic must be provided and registered there.

1.3.1 SKIDATA Staging environment

1. Get integrator Basic Authentication for SKIDATA Sandbox
2. Implement the client for the [OpenAPI Definition for swweb® Interface 'Dynamic Data' \(REST\) API](#)
3. Implement the server for the [OpenAPI Definition for swweb® Interface 'Dynamic Data' \(REST\) Event API](#)

- a. Start with HTTP
 - b. Implement the endpoints /v1/postEvent and /v1/health
4. Get SKIDATA to configure the URL and Basic Auth credentials of the server to receive push events
5. Enable the push events for a valid list of carparks, see [Enable Push Event Notification](#)
6. Trigger resending of current status, see [Trigger resending all counters of a Parking Facility](#)
7. Switch to HTTPS

1.3.2 SKIDATA Production environment

1. Get integrator Basic Authentication for SKIDATA Production
2. Get SKIDATA to configure the URL and Basic Auth credentials of the server to receive push events
 - a. On Production HTTPS is to be used
3. Enable the push events for a valid list of carparks, see [Enable Push Event Notification](#)
4. Trigger resending of current status, see [Trigger resending all counters of a Parking Facility](#)

2 Authentication

Basic Authentication is used, the username and password is provided by SKIDATA. Example:

```
curl --location --request GET 'https://sweb.skidata.com/bei/advconn/dynamicdata/v1/health' \
--header 'Authorization: Basic YWRtaW46cGFzc3dvcmQ='
```


3 Versioning

Version is defined in the resource URL, <https://sweb.skidata.com/bei/adv-conn/dynamicdata/v1/health/..> Backward compatibility is provided for older versions at least up until discontinuation thereof, discontinuation is announced up front by SKIDATA well before the actual date thereof.

4 Setup

4.1 URLs

4.1.1 Production URLs

- EU Car: <https://car.webhost.skidata.com/bei/advconn/dynamicdata>
- ASIA PACIFIC: <https://car.au.skidata.com/bei/advconn/dynamicdata>
- US: US US East <https://hb.usa.skidata.com/bei/advconn/dynamicdata>

4.1.2 Staging URLs

- <https://sweb.skidata.com/bei/advconn/dynamicdata>

5 API Definitions

5.1 OpenAPI Definition for sweb® Interface 'Dynamic Data' (REST) API

The interface description is based on OpenAPI v3, see <https://swagger.io/specification/>

[Interface definition](#)

Human readable documentation:

- [HTML description](#)

5.2 OpenAPI Definition for sweb® Interface 'Dynamic Data' (REST) Event API

The API has to be implemented by the integrator and enabled/configured by SKIDATA.

[Interface definition](#)

Human readable documentation:

- [HTML description](#)

6 Use Cases

6.1 Use Cases for OpenAPI Definition for sweb® Interface 'Dynamic Data' (REST) API

6.1.1 Fetch Counting Categories and Areas

Note:

The calls does not return the current counting levels and is not a pre-requisite for the event notation. However the returned list states the car parks that are required to enable the event push event notification, see [Enable Push Event Notification](#).

Retrieve a list of all defined Counting Categories or Areas respectively.

The complete lists of Counting Categories and Areas can be fetched regularly like on a daily basis.

The [OpenAPI Definition for sweb® Interface 'Dynamic Data' \(REST\) Event API](#) should be implemented by the integrator to receive updates as the counting values change.

Example for v1/countingcategories:

```
GET 'https://sweb.skidata.com/bei/advconn/dynamicdata/v1/countingcategories/1750' \  
--header 'Accept: application/json' \  
--header 'Authorization: Basic U2Nvbkr1bw9JbnRlZ3JhdG9yXzQ2NTYzMzpwYXNzd29yZDEyMyE='
```

```
[  
  {  
    "carparkId": 0,  
    "countingCategoryId": 1,  
    "name": "Short-term",  
    "capacity": 351,  
    "occupancyLimit": 320,  
    "freeLimit": 300  
  },  
  {  
    "carparkId": 0,  
    "countingCategoryId": 2,  
    "name": "Contract",  
    "capacity": 9999,  
    "occupancyLimit": 9999,  
    "freeLimit": 9999  
  },  
  {  
    "carparkId": 0,  
    "countingCategoryId": 3,  
    "name": "Total",  
    "capacity": 9999,  
    "occupancyLimit": 9999,  
    "freeLimit": 9999  
  }  
]
```

```
    "occupancyLimit": 9999,  
    "freeLimit": 9999  
  }  
]
```

6.1.2 Enable Push Event Notification

Enables push events for Parking.Logic Facility 26, carpark numbers 0, 1, 2 and 3. The operation is idempotent. If an existing configuration exists, e.g. carpark 4 is already enabled, the result will be that 0, 1, 2, 3 and 4 are enabled.

```
PUT 'https://sweb.skidata.com/bei/advconn/dynamicdata/v1/notifications/enable/26' \  
--header 'Content-Type: application/json' \  
--header 'Accept: application/text' \  
--header 'Authorization: Basic SW5qZWNoVG1ja2V0VGZzdF84MTIwNzQ6cGFzc3dvcmQxMjMh' \  
--data-raw '[  
  0,1,2,3  
]'
```

6.1.3 Disable Push Event Notifications

Disables push events for Parking.Logic Facility 26, carpark numbers 0, 1, 2 and 3. The operation is idempotent. If values are provided that are not previously configured, there are no side effects.

```
PUT 'https://sweb.skidata.com/bei/advconn/dynamicdata/v1/notifications/disable/26' \  
--header 'Content-Type: application/json' \  
--header 'Accept: application/text' \  
--header 'Authorization: Basic SW5qZWNoVG1ja2V0VGZzdF84MTIwNzQ6cGFzc3dvcmQxMjMh' \  
--data-raw '[  
  0,1,2,3  
]'
```

6.1.4 Fetch the enabled carparks

To verify the enabled carparks:

```
GET 'https://sweb.skidata.com/bei/advconn/dynamicdata/v1/notifications/config/1376' \  
--header 'Accept: application/json' \  
--header 'Authorization: Basic Q2xpZW50U3lzdGVtVXNlc19BZHZDb25uOnNrawRhdGEzMjNDQ0'
```

Response example:

```
[  
  2,  
  3,  
  0,  
  183,  
  83,  
  1,  
  12
```

```
]
```

6.1.5 Fetch the integrator configuration

After the API to receive push events is integrated, the integrator system URL and basic authentication have to be configured by SKIDATA, see [Receive push events](#). To verify the configuration the endpoint `v1/notifications/config` is available:

```
GET 'https://sweb.skidata.com/bei/advconn/dynamicdata/v1/notifications/config' \  
--header 'Accept: application/json' \  
--header 'Authorization: Basic Q2xpZW50U3lzdGVtVXNlc19BZHZDb25uOnNraWRhdGEzMjMzNDQ0'
```

Response example:

```
{  
  "url": "https://example.integrator.com",  
  "user": "username"  
}
```

6.1.6 Trigger resending all counters of a Parking Facility

Pre-requisites:

- Implementation of the [OpenAPI Definition for sweb® Interface 'Dynamic Data' \(REST\) Event API](#) as described below.
- SKIDATA has configured the base URL for the integrator service
- [Enable Push Event Notification](#) has been done for the carparks for which to receive events.

Example

```
PUT 'https://sweb.skidata.com/bei/advconn/dynamicdata/v1/counters/1750/resend' \  
--header 'Accept: application/text' \  
--header 'Authorization: Basic U2Nvbkr1bW9JbnRlZ3JhdG9yXzQ2NTYzMzpwYXNzd29yZDEyMyE='
```

6.2 Receive push events with OpenAPI Definition for sweb® Interface 'Dynamic Data' (REST) Event API

Caution:

HTTPS should be used for push events on production system



The HTTPS-Traffic is only possible for ports 443, 563, 4443, 16303, 2195 or 5310 to 5380!

HTTP-Traffic to be used in the integration phase is possible for ports 80, 81, 1980, 8080, 8091, 55561.

**Note:**

The direction change from the integrator calling the SKIDATA sweb Interfaces to sweb Interfaces calling the integrator.

6.2.1 Health endpoint

**Note:**

Only used at configuration time (by SKIDATA) when the integrator system URL is configured to verify the endpoint. The content of the response is irrelevant, only the HTTP return code 200 is checked.

```
GET 'https://example.integrator.com/health'
```

6.2.2 postEvent endpoint

Given that an integrator service on <https://example.integrator.com> that is configured, the events will be sent to <https://example.integrator.com/v1/postEvent>.

The event payload received is formatted in JSON. The cloudevent headers(ce-*) refers to the type and schema(OpenAPI definition) of the payload, e.g.

- ce-dataschema: <https://nexus.skidata.net/nexus/repository/site/sweb-Interface-DynamicData-REST-API/doc/rest/dynamicdata.events/v1/dynamicdata.events.yaml>

In the example below, a CountingCategory update is received as described in the [OpenAPI Definition for sweb® Interface 'Dynamic Data' \(REST\) Event API OpenAPI definition](#).

```
POST https://example.integrator.com/v1/postEvent HTTP/1.1
Accept-Encoding: gzip, deflate
Authorization: Basic dXNlcm5hbWU6cGFzc3dvcmQ=
ce-dataschema: https://nexus.skidata.net/nexus/repository/site/sweb-Interface-DynamicData-REST-API/doc/rest/dynamicdata.events/v1/dynamicdata.events.yaml
ce-id: be1be322-5049-4849-9978-6b9c5117bad7
ce-source: http://skidata.com/cloudevents
ce-specversion: 1.0
ce-time: 2022-10-12T11:46:51.096Z
ce-type: com.skidata.bei.dynamicdata.counter.v1.countingcategory
Content-Type: application/json; charset=utf-8
RequestID: f9161419-d63e-44d2-a74f-2fc713018545
Content-Length: 257
Host: 192.168.26.52:8080
Connection: Keep-Alive

{"carpark":{"id":83,"shortName":"SHPAC","name":"SHORT PARKING CATEGORY","facilityNr":1376},
"countingCategoryId":16,"name":null,"level":30,"capacity":89,"occupancyLimit":50,
```

```
"freeLimit":70,"externalCounting":false,"trafficSignalMode":0,"trafficSignalState":1}
```

**Note:**

We strongly recommend to use the provided SDKs from <https://cloudevents.io/> where possible.

6.2.2.1 Supported events

6.2.2.1.1 com.skidata.bei.dynamicdata.counter.v1.countingcategory

```
.....  
ce-type: com.skidata.bei.dynamicdata.counter.v1.countingcategory  
ce-dataschema: https://nexus.skidata.net/nexus/repository/site/swab-Interface-  
DynamicData-REST-API/doc/rest/dynamicdata.events/v1/dynamicdata.events.yaml  
.....
```

```
{  
  "carpark": {  
    "id": 83,  
    "shortName": "SHPAC",  
    "name": "SHORT PARKING CATEGORY",  
    "facilityNr": 1376  
  },  
  "countingCategoryId": 16,  
  "name": null,  
  "level": 30,  
  "capacity": 89,  
  "occupancyLimit": 50,  
  "freeLimit": 70,  
  "externalCounting": false,  
  "trafficSignalMode": 0,  
  "trafficSignalState": 1  
}
```

6.2.2.1.2 com.skidata.bei.dynamicdata.counter.v1.countingarea

```
ce-type: com.skidata.bei.dynamicdata.counter.v1.countingarea  
ce-dataschema: https://nexus.skidata.net/nexus/repository/site/swab-Interface-  
DynamicData-REST-API/doc/rest/dynamicdata.events/v1/dynamicdata.events.yaml
```

Example:

```
{  
  "carpark": {  
    "id": 83,  
    "shortName": "SHPAA",  
    "name": "SHORT PARKING AREA",  
    "facilityNr": 1376  
  },  
  "countingAreaId": 6,  
  "name": null,  
  "level": 50,  
}
```



```
"capacity": 89,  
"occupancyLimit": 50,  
"freeLimit": 70,  
"externalCounting": false,  
"trafficSignalMode": 0,  
"trafficSignalState": 1  
}
```